WorkTrace: How Coding Agents Explore, Change Code, and Check Their Work

Anonymous working manuscript

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Abstract

Coding agents inspect files, change code, and run checks. Final success and tool-call counts do not show how agents organize this work. We present WorkTrace, a four-stage framework for this analysis. It links each tool operation to a work target, an inferred code version, and earlier evidence. We study 6,300 runs from three models under one harness. We also study 6,000 runs from one model under three harnesses. The analysis shows two work patterns. First, Gemini performs more checks on the final code version than on earlier versions. This gap remains under three saved version rules. Second, OpenHands and SWE-agent revisit known targets more often than mini-swe-agent on the same tasks. Their mean revisit rates are 32.9%, 30.0%, and 18.2%, respectively. Exact-repeat rates are close to 0.2% for all three harnesses. These findings show where work accumulates and which system comparisons need closer inspection. They do not establish that the additional work is unnecessary. We explain how evidence about test scope and unresolved questions can support that next assessment.

Working-draft status. This revision adds statistical analyses of saved data. It adds no agent runs or human labels. Red notes identify the next research steps.

REVISION 01. Repair the state-tracking errors before treating these results as final. The new analyses test saved measurements. They do not repair those measurements.

1 Introduction

Coding agents often solve the same task through different sequences of work. One agent may inspect many files before an edit. Another may edit early and then run many checks. A final patch and a tool-call count cannot explain this difference.

We need to know where the work occurs. Does an agent check each code version, or mainly the final version? Does a harness return to known targets, even when it rarely repeats an exact command and output? These questions help researchers compare systems and choose cases for closer analysis.

TRACE provides a useful approach to such questions (Zhang et al., 2026). It first measures broad behavior, then reconstructs reasoning graphs, and then studies common patterns. It connects those patterns to intermediate answer correctness and further utility tests. Coding-agent traces require a different representation. Code changes during a run, and two different commands can test the same code property.

We introduce WorkTrace to reconstruct this external work. The framework has four stages: collect operations, infer work relations, construct progression graphs, and compare graph patterns. Two linked views show target visits and code versions. Together, they connect exploration, code changes, and verification.

Our study has two comparison sets. The first compares three models on the same tasks under mini-swe-agent. The second compares three harnesses on the same tasks with Qwen fixed. The results support two findings:

- **Final-version checking.** Gemini places more checks on its final inferred code version than on an average earlier version. Luna and Terra show no similar positive gap.
- **Target revisits.** OpenHands and SWE-agent return to known targets more often than mini-swe-agent. This difference remains when we use the same rate definition and match tasks.

The framework makes these patterns measurable. It also identifies the evidence needed to interpret them. More checks can improve coverage. A file revisit can answer a new question. We therefore separate an observed work pattern from a claim that the work is excessive.

REVISION 02. Keep these two findings central. Explain the decision each finding supports. Confirm the patterns on fresh tasks after the measurement repairs.

2 Related Work

Work and reasoning structures. TRACE studies how proposed answers change during reasoning (Zhang et al., 2026). Bouzenia and Pradel study thought–action–result sequences in software agents (Bouzenia and Pradel, 2025). WorkTrace studies targets, code versions, and evidence within the same run. These links support comparisons of where agents spend their work.

Redundant work and efficiency. RedundancyBench evaluates redundant steps and their supporting evidence (Zhu et al., 2026). Its annotation procedures can inform our validation. AgentDiet reduces agent history (Xiao et al., 2026). TRIM studies unnecessary changes in final patches (Mathai et al., 2026). These studies address related questions, but each uses a different outcome. A repeated observation, a removable history item, and an unnecessary code change are not interchangeable labels.

Interaction with users. SWE-TaskFlow studies how benchmark task delivery matches user task flows (Slinko et al., 2026). Such data can test how new requirements change an agent's work. Our current comparisons use fixed task instructions.

REVISION 03. Credit any annotation procedure adapted from RedundancyBench. Add an interaction study only if it tests an explanation of the two main findings.

3 Benchmark Agent Work

We first compare final success and the amount of work. We then use WorkTrace to locate that work within each run.

3.1 Across Models

The first set contains 700 tasks, three models, and three runs per task and model. All 6,300 runs use mini-swe-agent. The saved operation table contains 188,609 operations. We retain the model names in the source metadata: Gemini 3.7 Flash, GPT-5.6 Luna, and GPT-5.6 Terra. Gemini has higher final success and uses more calls than either GPT model (Table 1). This comparison does not show whether its additional work is useful. The models and their released settings differ. They also differ in the number of code versions and checks, which we study in Section 5.

3.2 Across Harnesses

The second set uses Qwen3.8-27B from Open-SWE-Traces v1.2 (NVIDIA, n.d.). It contains 2,000 shared tasks with one selected run per harness. The three harnesses are mini-swe-agent, OpenHands, and SWE-agent. Together, the 6,000 runs contain 370,992 operations. OpenHands and SWE-agent use more calls than mini-swe-agent (Table 2). All three have exact-repeat rates near 0.2%. However, their target revisit rates differ. Section 5.1 examines this difference with paired task comparisons.

3.3 Comparison Method

We analyze the model and harness sets separately because they contain different tasks. These sets cannot establish a model–harness interaction.

The additional analyses use 5,000 task-bootstrap samples with seed 20261026. Each sample draws tasks with replacement and keeps their associated runs together. Paired comparisons keep the same task on both sides. We report 95% percentile intervals. These are exploratory intervals, without adjustment for multiple comparisons.

For the version analysis, we require at least two inferred versions and a final version other than the initial repository state. We then select task–repeat pairs that meet this rule for every model under every version rule. This gives 892 matched pairs from 418 tasks, with 892 runs per model. This selected subset tests whether the pattern persists under common conditions. It does not represent every run.

REVISION 04. Report all exclusions and settings in the final release. The saved metadata includes 4,198 missing results. Identify their causes before claiming complete observations.

4 Framework for Work Progression

WorkTrace maps a tool sequence into a description of the work. Table 3 defines the main terms.

4.1 Stage 1: Collect Operations

We use saved tool traces from the two comparison sets. The parser records each call, its result, its position, and the available exit status. It assigns operation types such as read, search, edit, run, and submit.

The parser also splits shell chains to inspect their parts. The native call remains the measurement unit. A call can contain several action types. Normalization removes selected volatile fields from execution outputs. Exact-repeat detection compares the normalized command and output hash.

4.2 Stage 2: Infer Work Relations

For each operation, the framework retrieves two possible earlier matches. One is the latest operation on the same target. The other is the latest same-type operation with the same usable output. The current rules exclude short or explicitly truncated outputs from the second search.

Rule facts describe whether the output matches and whether code may have changed. An automatic annotator then labels the operation's role, relation, and contribution. Roles include exploration, revision, and verification. Relations include unmatched, revisit, reproduction, and uncertain. Contributions can be distinct, non-distinct, or uncertain.

The validator requires a cited match for a non-distinct label. Thus, the labels measure contribution within the retrieved evidence. They do not measure all unnecessary work. Appendix A reports the current annotation checks.

REVISION 05. Test candidate recall with independent readers who can inspect any earlier operation. The current rules prevent unmatched operations from receiving a non-distinct label.

4.3 Stage 3: Construct Progression Graphs

Target graph. Each node represents a work target. Directed edges follow the operation sequence. Consecutive visits to one target form a self-loop. A later return forms a return edge. These edges locate repeated attention without assuming repeated information.

For example, an agent can read two different ranges of one file. The target repeats, but the second range can supply new evidence. Target identity also depends on the tool interface. A structured editor may expose a file path directly. A shell command may encode that path within a more specific expression.

Version graph. Each node represents an inferred code version. Detected code changes create new versions.

Model	Runs	Success (%)	Calls/run	Input tokens (M)	Exact repeats (%)
Gemini 3.7 Flash	2,100	60.0	53.1	3.34	2.21
GPT-5.6 Luna	2,100	50.0	17.6	0.25	0.02
GPT-5.6 Terra	2,100	54.2	19.0	0.24	0.05

Table 1: Saved cross-model benchmark on 700 matched tasks. Tokens and calls are per-run means. Exact repeats require the same normalized command and output. Their rates are means of per-run rates. Success means that the dataset marks the task as resolved.

Harness	Runs	Success (%)	Calls/run	Exact repeats (%)	Target revisits (%)
mini-swe-agent	2,000	57.1	51.8	0.20	18.2
OpenHands	2,000	51.0	66.2	0.18	32.9
SWE-agent	2,000	54.5	67.5	0.20	30.0

Table 2: Cross-harness benchmark with Qwen fixed. This revision calculates all rates as means of per-run rates. Older saved summaries used pooled operation rates for revisits. The new values permit a consistent comparison.

Term	Meaning in this paper
Operation	One native tool call and its recorded result.
Target	The file, command, or tool object that an operation addresses.
Revisit	An operation with an earlier same-target candidate.
Version	An inferred code state after a detected change.
Check	A run operation attached to a version, excluding detected read-only inspections.
Non-distinct	The annotator finds no new contribution relative to the cited earlier evidence.

Table 3: Terms describe recorded work. A check is not automatically a test, and a revisit is not automatically redundant.

Checks attach as self-loops to the version on which they run. Recognized restores can return to earlier versions.

A new node after a failed check receives the stored rule label `correct`. This label means revision after failure. It does not mean the resulting code is correct. Only final task success has a dataset outcome.

We retain three saved version rules. The first creates a version for every detected write, including uncertain writes. The second uses known writes only. The third excludes writes confined to inferred support files. Support files include temporary diagnostic scripts. The third rule uses final submission metadata, so it is a retrospective analysis.

REVISION 06. Fix false state matches, branch changes, unknown side effects, failed restores, and incomplete diffs. Rebuild the graphs before making final claims about version counts or backtracks.

4.4 Stage 4: Compare Patterns

We group version graphs by model and version count. New versions advance horizontally. A new path after a backtrack moves to a higher row. Graphs align by these coordinates.

Node and edge weights record how often each component appears. The saved figures retain components present in at least 30% of grouped runs. We display groups with at least 20 runs. This procedure follows TRACE's coordinate-based aggregation approach (Zhang et al., 2026).

We also calculate measures before graph pruning. The verification gap compares final-version checks with the mean checks on earlier versions. The revisit rate divides operations with a same-target candidate by all operations in a run. These measures support direct comparisons beyond visual inspection.

REVISION 07. Test the graph alignment and pruning thresholds. A simple-looking aggregate can hide uncommon paths. Use unpruned run measures for numerical findings.

5 Findings

5.1 Work Progression Patterns

TRACE's Explorer and Late Landing patterns depend on intermediate answer correctness (Zhang et al., 2026). Our data lacks intermediate code-correctness labels. We therefore name our patterns by the work that we observe.

Pattern I: Final-version checking. Gemini performs more checks on its final inferred version than on an average earlier version. Figure 2 shows this pattern in the saved four-version groups. We test it numerically on the common subset defined in Section 3.3. For run r, we define the verification gap as

$$G_r = C_{r,\text{final}} - \frac{1}{|V_r^-|} \sum_{v \in V_r^-} C_{r,v}, \tag{1}$$

where $C_{r,v}$ counts checks on version v. The set V_r^- contains other inferred versions and excludes the initial repository state. A positive gap means more checks on the final version.

Under the support-file exclusion rule, Gemini averages 3.15 final checks and 0.61 checks per earlier version. Luna averages 0.39 and 0.43. Terra averages 0.47 and 0.57. These values use the same 892 task–repeat pairs per model.

Gemini's mean gap exceeds Luna's by 2.58 checks [2.30, 2.84]. It exceeds Terra's by 2.64 checks [2.37, 2.92]. The direction remains under the other two version rules (Figure 3a). The Gemini–Luna contrast ranges from 1.80 to 2.58 checks across these rules. The gap also appears

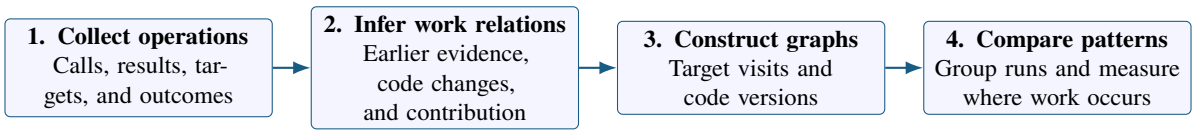


Figure 1: Four stages of WorkTrace. Operation IDs connect both graph views to the source records. The graphs summarize external actions. They do not reveal hidden model reasoning.

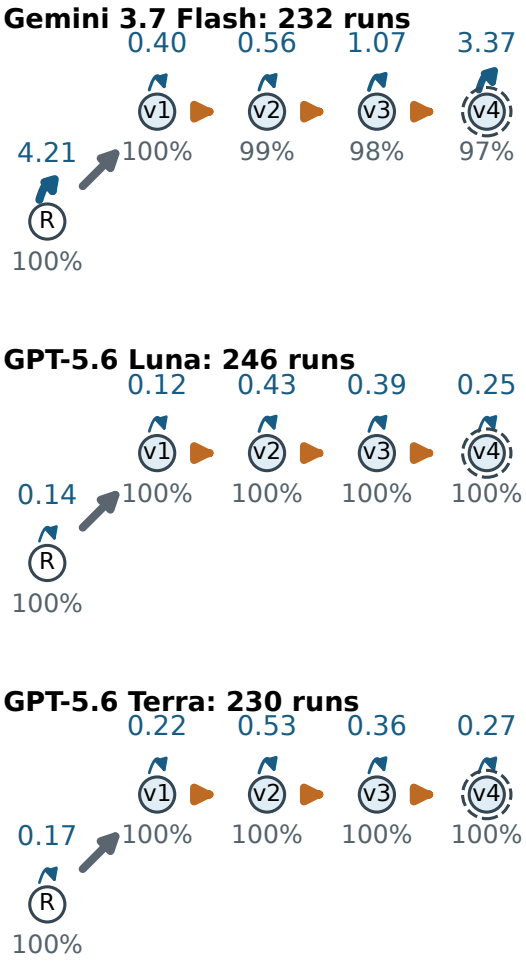


Figure 2: Saved four-version patterns. R is the initial repository state. Blue loops show mean checks at each coordinate. Percentages show node presence. Dashed circles mark frequently final nodes. Rare components are pruned. These graphs do not mark correctness.

in both resolved and unresolved Gemini runs. Under support-file exclusion, their mean gaps are 2.46 [2.19, 2.73] and 2.48 [2.12, 2.85], respectively. These groups use all eligible Gemini runs, rather than the common subset. The association therefore does not require final task success.

What this explains. Gemini's additional execution work includes a final-version checking phase. Total call counts cannot locate that phase. The graph can. However, the final version can fail the task. Its position does not establish correctness or sufficient verification.

REVISION 08. Identify which properties each check tests. Compare new coverage, repeated coverage, and stochastic reruns. Measure the time available for checks before interpreting the gap as excessive verification.

Pattern II: Target revisits. With Qwen fixed, the harnesses differ in how often they return to known targets. Mean revisit rates are 18.2% for mini-swe-agent, 32.9% for OpenHands, and 30.0% for SWE-agent. These use one consistent per-run definition (Figure 3b).

Paired comparisons give an OpenHands–mini-swe-agent gap of 14.73 percentage points [14.26, 15.19]. The SWE-agent–mini-swe-agent gap is 11.81 points [11.37, 12.25]. Removing the task with 164 repeated probe executions changes these gaps to 14.76 and 11.85 points. The direction also holds within both source datasets (Appendix B).

Exact-repeat contrasts are much smaller. Relative to mini-swe-agent, the mean differences are −0.013 points for OpenHands and +0.004 points for SWE-agent. Their intervals include zero. This does not establish equivalence, but it shows that exact repeats alone miss the much larger revisit contrast.

What this explains. The harnesses organize recorded target access differently. They can revisit a file through a new command or a new range. Tool interfaces also change how the parser identifies targets. Thus, the pattern identifies a comparison that requires evidence inspection. It does not prove memory failure or excessive exploration.

REVISION 09. Normalize file paths, read ranges, and mixed calls across harnesses. Then inspect whether the remaining revisits resolve new questions or return to questions already settled.

5.2 Trace the Evidence

A graph shows where to inspect. Evidence about an operation shows what we can conclude from it.

Do later final-version checks use new commands? The saved Gemini reconciliation contains 6,699 final-version checks across 2,100 runs. A run's first check is automatically a new command within that version. We remove this first check from each nonempty run before assessing later checks.

The remaining 4,964 checks include 89.0% new commands [87.5, 90.5]. Identical command-and-output reruns account for 2.6% [2.1, 3.1]. Same-command checks with different outputs account for 8.4% [7.1, 9.8]. These rates pool checks and resample tasks.

Thus, most later checks are not exact reruns. But command variety does not establish new test coverage. Two pytest commands can select overlapping tests. Different tests can exercise the same property. This evidence supports a checking phase, while leaving its benefit unresolved.

REVISION 10. Extract test IDs, assertions, configuration, and relevant code changes. Replace command variety with a measure of the evidence each check adds.

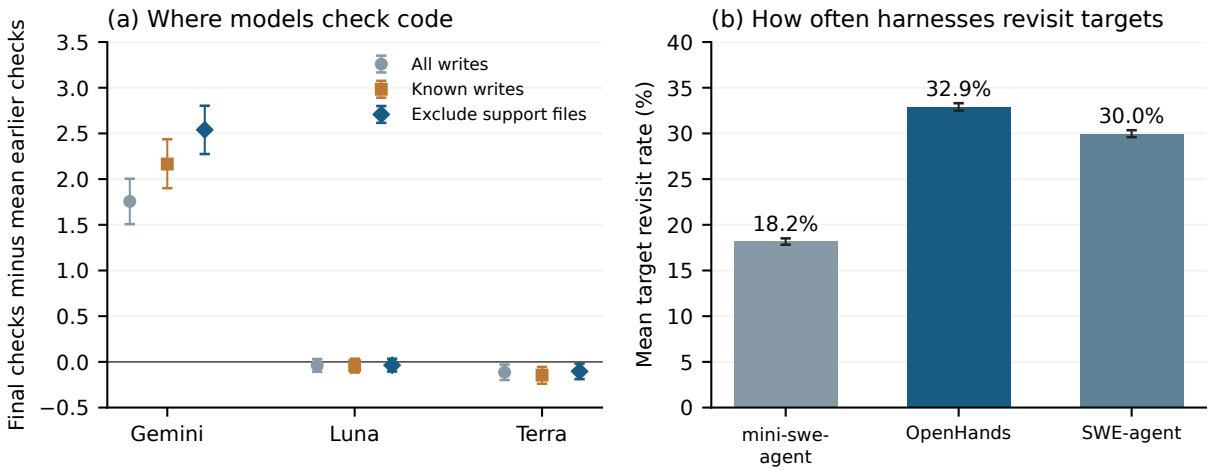


Figure 3: New analyses of saved data. (a) Verification gaps under three version rules on a fixed subset of 418 tasks. Each model has 892 runs. (b) Mean revisit rates on the same 2,000 tasks for each harness. Bars and points use 95% task-bootstrap intervals. Both plots describe the current measurements, before state-tracking repairs.

Do revisits add new information? The 50-run model pilot contains 1,432 operations. Of 225 operations with retrieved candidates, 51 receive a non-distinct label. Only 21 receive both reproduction and non-distinct labels. Those 21 are candidates for redundancy under the current evidence rules.

The 60-run Qwen pilot contains 4,018 operations. Non-distinct shares among operations with candidates are 5.1% for mini-swe-agent, 5.2% for OpenHands, and 9.7% for SWE-agent. These small pilot estimates cannot establish the meaning of every revisit in the 6,000-run set.

The important next distinction is whether a revisit addresses an unresolved question. A return can confirm a code change, inspect another file range, or recover missing context. Each case requires the operation and its earlier evidence. Graph shape alone cannot decide which case applies.

REVISION 11. Build an independent human reference for matched and unmatched operations. Include new ranges, relevant code changes, missing outputs, and valid matches outside the retrieved candidates.

5.3 Towards Managing Excess Work

We use *excessive work* to mean continued work whose expected benefit is too small for its cost, under a stated decision rule. This definition requires an outcome and a cost. Counts and loops provide candidates for assessment.

Verification. A possible warning is repeated checking on an unchanged relevant state with no new coverage. The relevant state includes code, test selection, configuration, and external conditions. A different command is not sufficient evidence of a new property. An identical rerun can still help when execution is stochastic.

Exploration. A possible warning is a return to a settled question without new evidence or changed requirements. Reading an ultimately irrelevant file can still eliminate a plausible cause. We must assess what the agent could know at the time.

Use of the framework. The version view locates checking phases. The target view locates repeated attention. Evidence links then distinguish new information from repeated information. This sequence gives a clear use for the framework: choose comparable cases, inspect the reason for the difference, and test the proposed explanation.

REVISION 12. Test this use directly. Ask independent readers to diagnose the same runs with simple summaries and with WorkTrace. Compare their accuracy on independently judged cases.

Next experiment. A controlled continuation experiment can test whether a warning identifies removable work. At a selected prefix, compare continued work with a suitable alternative. Measure final success and total cost, including analysis overhead. Removing recorded operations while keeping the recorded suffix fixed is not an adaptive agent experiment.

REVISION 13. No stopping or work-removal result is available. Set the success and cost criteria before the experiment. A nonsignificant success difference does not prove preserved performance.

6 Conclusion

WorkTrace reconstructs how coding agents inspect targets, change code, and check versions. The current data shows two patterns: Gemini concentrates more checks on the final inferred version, and the Qwen harnesses differ in target revisits. New paired analyses preserve these differences under the tested measurement choices.

These patterns make system comparisons more specific. They identify where work accumulates and which evidence needs inspection. Independent validation and corrected state tracking remain necessary before using the framework to measure excessive work.

REVISION 14. After validation, rewrite the main claim around what the framework explains. Keep the two findings and their evidence clear. Do not transfer TRACE's correctness-based conclusions without an intermediate correctness measure.

Limitations

Measurement. The audit reproduces errors in state identity, branch changes, mutation targets, restore outcomes, and incomplete diffs. These errors can change the version graphs. The new sensitivity analyses use existing measurements and cannot establish that the errors are harmless.

Validation. Automatic annotators agree on many labels, but the intended human reference remains incomplete. Candidate retrieval restricts the labels. The external CodeTraceBench result is unfavorable and appears in Appendix A.

Scope. Version matching selects runs with particular observed behavior. Target identifiers depend on tool interfaces. The two comparison sets use different tasks and released configurations. We therefore report system associations rather than isolated model or harness causes.

Utility. We do not observe intermediate patch correctness or the outcomes of stopping early. We also lack a complete monetary or time cost for every operation. Additional checks and revisits must not be counted as wasted work by default.

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A Validation

Automatic annotation. The primary annotator is recorded as DeepSeek-V4.1-Flash through the configured endpoint. The JetBrains pilot has 50 runs from 17 tasks. The Qwen pilot has 60 runs from 20 tasks. Larger partial label files are excluded from these pilot results.

A secondary annotator, GLM-5.3-Flash, labels 225 matched items and 200 sampled unmatched items from the JetBrains pilot. Matched contribution agreement is 90.2%, with $\kappa = 0.70$. Matched relation agreement is 84.0%, with $\kappa = 0.50$. Candidate-redundancy agreement across 425 items has $\kappa = 0.83$. Three primary-model samples give contribution $\kappa = 0.88$–0.90 on matched items.

These checks measure agreement. They do not establish accuracy. Both annotators receive the same candidates and rule facts. The 200-item human packet has no adjudicated contribution labels.

External comparison. The saved CodeTraceBench analysis includes 484 trajectories, 17,317 steps, and 69 human-labeled unuseful steps (NJU-LINK, n.d.). Table 4 reports the result. Only 32 of the 69 human-labeled steps have a retrieved WorkTrace candidate.

Predictor	Predicted	Precision	Recall
Exact repeat	412	5.6%	33.3%
Target revisit	4,599	0.6%	39.1%
Candidate redundant	536	3.5%	27.5%

Table 4: Saved comparison against unuseful-step labels. These labels differ from WorkTrace's contribution construct. The result still provides no positive external validation.

Return after a detour. The saved pilot includes 153 returns after visits to other targets. Removing two uncertain contributions leaves 151 returns, 32 non-distinct labels, and 17 task clusters. A task-clustered GEE gives adjusted odds ratios of 32.7 [2.3, 456] for two to three intervening targets. The estimate for four or more is 14.5 [1.1, 193], relative to one target. The model controls run position, run length, and model identity. The intervals are wide, and the pattern mainly occurs in long runs. We retain this as a secondary exploratory result.

B Additional Analyses

Version sensitivity. Table 5 uses the same 892 task–repeat pairs under all three rules. Every model has 892

runs from 418 tasks. Each contrast subtracts Luna's within-run verification gap from Gemini's gap.

Version rule	Difference	95% interval
All writes	1.80	[1.54, 2.05]
Known writes only	2.21	[1.94, 2.48]
Exclude support files	2.58	[2.30, 2.84]

Table 5: Paired Gemini–Luna contrasts in checks. Sensitivity to version rules changes the size but not the direction. All rules share unresolved implementation risks.

Source-dataset sensitivity. The matched harness set contains 1,063 Scale-SWE tasks and 937 SWE-rebench-V2 tasks. Table 6 compares each harness with mini-swe-agent within a source dataset.

Source	Harness	Gap	95% interval
Scale-SWE	OpenHands	16.60	[15.96, 17.23]
Scale-SWE	SWE-agent	11.80	[11.17, 12.39]
SWE-rebench	OpenHands	12.60	[11.93, 13.27]
SWE-rebench	SWE-agent	11.82	[11.21, 12.45]

Table 6: Paired differences in mean revisit rates, in percentage points. The direction holds within both sources under the current target definition.

Exploration and uncertainty. These are additional analyses of data already used to develop the framework. Their intervals describe sampling variation under the stored measurements. They do not include parser error, annotator error, or the effect of selecting findings after inspection. Fresh tasks are required for confirmation.

C Data and Reproduction

Saved inputs. The model benchmark uses `results/v2/benchmark/table1_by_model.csv`. The new harness analysis uses `results/v2/task5/runs_benchmark.parquet`. Version analyses use the three saved `runs_all.csv` files under `results/v2/viewb/`. The final-check analysis uses `results/v2/reconcile/scale_per_run.csv`.

Pilot results use `results/v2/task2/`, `results/v2/task3/`, `results/v2/task4/`, and `results/v2/qwen/pilot/`. The figure script redraws the saved four-version aggregates. It does not infer new graphs.

New outputs. The script `paper/worktrace-draft/analyze_saved_data.py` writes to `results/paper_revision_20261026/`. It produces paired contrasts, version sensitivity tables, outcome strata, source strata, and final-check categories. Assertions check sample sizes, matching, and count identities. The output manifest records the seed, package versions, and input hashes.

Dataset versions. JetBrains records revision `a5ec6ad84287`. Open-SWE records `a85dc97891bd`. The manuscript manifest retains the full hashes. Raw traces and some earlier caches are absent from this checkout.

REVISION 15. Release the full input-to-output map with parser, prompt, and dataset versions. Preserve complete observations so others can inspect the evidence behind each label.

D One-Week Revision Priorities

Days 1–2: Repair measurements. Fix the seven reproduced audit cases. Rebuild state facts and version graphs. Report how the fixes change both main findings.

Days 3–4: Validate evidence. Prepare an independent reference across models and harnesses. Include unmatched operations and difficult cross-tool cases. Keep uncertainty when the trace cannot support a judgment.

Days 5–6: Test explanations. Measure test scope for final-version checks. Normalize target access across harnesses. Compare simple summaries with the full framework on independently judged cases.

Day 7: Consolidate the study. Update the figures and main claims. Define a fresh-task confirmation set. Add a continuation experiment only if the required environment and budget are available.

REVISION 16. Prioritize reliable measurements and a direct test of the framework's value. More labels or a larger graph gallery alone will not resolve the main weaknesses.